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COMMERCIAL ITEM DESCRIPTION

# FREEZER, MECHANICAL, BLOOD PRODUCTS

## Ultra Low Temperature, Chest Type, 500 Unit Capacity

The General Services Administration has authorized the use of this Commercial Item Description for all Federal Agencies.

1. SCOPE. This Commercial Item Description covers an ultra low temperature, blood products freezer.

2. Salient Characteristics.

2.1 Freezer type and application. The freezer shall be a mechanically operated, air cooled, chest type, ultra low temperature freezer, suitable for freezing and maintaining blood products at -121°F(-85°C), including red blood cells, plasma, platelets, and cryoprecipitates.

2.2 Freezer chamber capacity. The storage chamber of the freezer shall have sufficient capacity to accommodate and store 500 units of red blood cells or plasma packaged in standard approved plastic bags and corrugated boxes, flat side down, 7-1/8 inches by 5-5/8 inches by 1-7/8 inches (filled and bulging dimensions).

Beneficial comments, recommendations, additions, deletions, clarifications, etc. and any other data which may be used to improve this document should be sent to: Defense Personnel Support Center, Directorate of Medical Materiel, ATTN: DPSC-MBP, 2800 South 20th Street, Philadelphia, PA 19145-0530, by letter or using the self-addressed Standardization Document Improvement Proposal, (DD Form 1426) appearing at the end of this document.

AMSC N/A FSC 4110

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2.3 Freezer design. The freezer shall include at least the following: chamber and outer walls and bottoms, insulation in-between, insulated lid with two (2) locking rings, inner sliding doors, two (2) complete refrigeration systems, liquid nitrogen back-up system (if desired by customer), anti-icing capability, refrigeration machinery compartment(s), air filter(s), chamber drain, fork lifting and floor mounting capability, and control panel. Provided on the control panel shall be controls, alarms, indicators, and a temperature recorder.

2.4 Construction material and finish.

2.4.1 The interior and exterior of the freezer shall be fabricated of stainless, or protected corrosion and rust resistant zinc finished steel plate having a minimum thickness of 22 gauge. Where painting or plating is used, commercially accepted practice shall be used for cleaning and priming prior to finishing. Finish shall provide a pleasing professional appearance and be free of gaps, scratches, chips and other imperfections.

2.4.2 The lid, walls, and bottom of the freezer shall be insulated with rigid polyurethane foam, foamed-in-place under pressure in a heated foaming station. The insulation thickness shall be 3-1/2 inches in the lid, 5 inches in the side walls and bottom, with a tolerance of ±3/16 inches for lid, walls and bottom. Wiring, sensors, or devices which may require repair or maintenance, shall not be placed inside the foam or be covered by the foam.

2.5 Freezer lid and locking rings. The freezer lid shall be counter balanced or strutted to hold any position from closed to fully opened with a maximum movement force of 10-15 pounds in either direction. The gaskets shall be of the magnetically sealing type. They shall be constructed with material which remains fully functional at the expected temperatures of use. Two locking rings shall be provided to enable locking of the lid. These locking rings shall be located on the front, from 1/4 to 1/3 of the freezer lid length, from each end of the lid.

2.6 Interior door lids. The chamber shall have a minimum of three movable or removable sublids constructed of Lexan or equal. Sublids shall allow access to part of the contents without exposing the entire contents to ambient conditions when the main lid is opened.



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2.7 Refrigeration systems.

2.7.1 The freezer shall be equipped with two complete, independent, cascaded, air cooled, refrigeration systems (4 motor/compressor combinations). Each cascaded system shall contain a high stage and a low stage. The evaporator of each low stage refrigeration system shall contain its own copper tube circuit, cooling all four sides of the chamber (and the bottom of the chamber if copper tube circuit is also used on the bottom). Each circuit shall be designed to uniformly cool the chamber; therefore, regardless of which cascade refrigeration system is operating, the chamber shall be uniformly cooled. Each cascaded system shall have an "on/off" switch. A key switch shall also be provided to enable start-up of refrigeration systems.

2.7.2 The four freezer compressors shall be of the reciprocating accessible semehermetic type; each compressor shall be rated a minimum of one horsepower. Oil level sight glasses and brass service valves shall be provided in order to visually observe oil level, and facilitate the use of gauges and the addition of refrigerants during servicing. Non-flammable refrigerants shall be used in the refrigeration systems. The four motors driving the compressors shall operate at 1750 rpm and shall be rated at least one horsepower each.

2.7.3 The refrigeration system shall have a minimum net thermal capacity of 3,830 BTU/HR empty at -121°F(-85°C) with both cascade systems operating in a temperature of 90°F(32°C).

2.7.4 The refrigeration system's cooling design shall be independent air cooled cascades to allow sufficient reserve capacity to maintain an internal chamber temperature as specified herein.

2.7.5 The total refrigeration system shall have the capability to maintain an internal chamber temperature of at least -121°F (-85°C), ±2°F/±1C° (both refrigeration systems operating), with ambient air temperature 110°F(43°C) ±2°F.

2.8 Back-up system. The freezer shall be provided with a liquid nitrogen back-up system if desired by customer. This system shall automatically initiate cooling in the event of compressor system failure or power failure. Freezer chamber shall be maintained at the operating temperature automatically for at least 24 hours with the use of this backup system. Electrical input shall be 208/230 volts, 50-60 Hz three phase. Batteries shall be provided as back-up to operate the system.



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2.9 Anti-icing. The unit shall be supplied with anti-icing to the extent that it will prevent freezer lock of the freezer door lids (outside door lid and inside sliding doors) and of the contents of the freezer when the unit is operated in a 95°F(35°C) environment, ±2°F 95 percent relative humidity ±5 percent at sea level atmospheric pressure. Opening the lid fully for 60 seconds (plus or minus 2 seconds), every 4 hours during a 24 hour period, shall produce no frost build-up or freezer-lock of doors (outside lid or inner sliding doors) or contents with a full, half-full, or empty freezer.

2.10 Machinery compartment. The refrigeration machinery compartment shall be designed to allow all servicing to be conducted from the front of the freezer.

2.11 Air filters. The air inlets to the refrigeration condensers shall be equipped with removable filters, capable of being cleaned with soap and water and reused continuously.

2.12 Chamber drain. A method shall be provided for draining the chamber from the bottom through a drain line piped to the front of the freezer and terminated in a 3/8 inch, female, pipe fitting with a removable plug for cleaning.

2.13 Temperature indicating devices.

2.13.1 A digital indicator shall be provided to indicate freezer chamber temperature. Means shall be provided to set the operating temperature within -85°F(-65°C) and -121°F(-85°C) (±2°F/±1°C). Indicated temperature shall be within ±5°F of absolute temperature. The indicator shall remain in operation even in the event of power failure.

2.13.2 The seven day recorder shall provide a continuous record of temperature with time. It shall continue to operate even through power failure. It shall be a single pen, inkless, pressure writing system with a seven day per revolution drive. The recording range shall include from 75°F(24°C) through -153°F (-103°C) temperature accuracy shall be within ±2°F/±1°C. It shall be easily visible and accessible for both temperature observation and for changing charts.

2.14 Alarm systems. The freezers shall be provided with line powered (208/230 volts, 50-60 Hz) and back-up battery powered (memory free) audible and visual alarms which shall activate when power fails or temperature deviations occur beyond the preset high and low temperature limits. Capability shall be provided for setting the high alarm limit within 6°F(-14°C) to 9°F(-13°C) above the operating temperature set point; the low alarm limit



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within 6°F(-13°C) to 9°F(-12°C) below the operating temperature. The operating temperature range shall be -85°F(-65°C) through -121°F(-85°C). The alarm set points shall be settable to within ±2°F/±1°C. The alarm shall also activate when the back-up cooling system is activated. The local alarms shall have the capability of being silenced temporarily with a push button or similar device, or being defeated with a key switch. Capability shall be provided for connection of the alarm circuitry to a remote alarm or a computerized central alarm/monitor system. The alarm systems shall continue to function for a minimum of 5 hours even when AC power is lost. The alarms shall be provided with a test mechanism which shall allow testing of alarm circuits for proper functioning.

2.15 Control panel. The freezer shall be provided with a control panel at the front. Included on this control panel shall be at least the following items:

1. Key operated "on/off" switch for refrigeration systems.
2. On/off switch for each cascaded refrigeration system.
3. Chart recorder which records chamber temperature.
4. Digital temperature indicator to show internal chamber temperature.
5. Chamber operating temperature setpoint mechanism.
6. Warm and cold limit alarm setpoint mechanisms.
7. Warm and cold setpoint temperature readout(s). (May use existing digital indicator with proper circuitry.)
8. Warm and cold limit alarms-audio and visual.
9. Power failure alarms-audio and visual.
10. Audio temporary silence button or mechanism.
11. Key operated audio alarm "on/off" switch.
12. Alarm test button or mechanism.
13. Remote alarm connectors.
14. Back-up refrigeration system power failure alarms-audio and visual.
15. Anticipating heater switch.

2.16 Fork lift. Fork lifting capability shall be built into the freezer chest to allow lifting of the freezer.

2.17 Mounting. A means to allow securing of the freezer to a floor, a ship deck, or an isocontainer, shall be provided in order to eliminate undesired movement.

2.18 Operating characteristics. The freezers shall be capable of maintaining operating temperatures at any operator selected temperature within the temperature range of -85°F(-65°C) through -121°F(-85°C). A mechanism shall be provided for setting the operating temperature within ±2°F/±1°C of desired temperature,



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with absolute accuracy of ±5°F. Operating temperature shall be controlled within ±5°F from readings at the sensing and control bulbs.

2.18.1 With operating temperature set at -121°F(-85°C) the compressors shall cycle at least once every 1/2 hour while internal temperature remains in the operating temperature range, ±6°F.

2.18.2 While the internal temperature of the freezer is maintained at -121°F(-85°C) there shall be no increase in motor current demand (measured at same point in cycle) when operated at an ambient temperature 95°F(+35°C) ±2°F.

2.18.3 In room ambient of 75°F(23°C), ±2°F each cascade refrigeration system shall be capable of maintaining an empty chamber at-121°F(-85°C); and each shall be capable of pull down from 72°F(22°C) to -121°F(-85°C) in less than four hours. Both systems simultaneously shall be capable of pull down from 72°F(22°C) to -121°F in less than two hours in the same ambient conditions.

2.18.4 Each or both refrigeration systems operating shall be capable of automatic, uninterrupted pull down to -121°F(-85°C) from 72°F(22°C).

2.18.5 The freezer shall be capable of maintaining -121°F (-85°C), at an ambient temperature up to 110°F(±2°F).

2.18.6 The freezer shall be capable of freezing a minimum of 27 boxes of simulated red blood cells, plasma, or platelets from 72°F(22°C) to -121°F(-85°C) in less than 16 hours.

2.18.7 The freezer shall be capable of reliable start-up after being shut off and left unattended for a minimum of one year at ambient temperature.

2.18.8 Normal operation shall permit the compressors to cycle on and off while maintaining the -121°F(-85°C) internal temperature, while the cascaded systems shall alternate from one system to the other. For rapid cool down of the contents of the freezer, means shall be provided to allow automatic simultaneous operation of both cascaded refrigeration systems. Such operation shall be effected automatically when the internal temperature of the freezer exceeds the warm alarm set point. Failure of either cascaded refrigeration system shall cause the surviving cascaded system to function and maintain the pre-set operating temperature indefinitely.



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2.19 Storage rack. The storage rack system shall be provided to facilitating orderly storage inventory requirements and selective unit recovery.

2.20 Electrical characteristics. The freezer shall operate over the full tolerance range of the following line voltage and frequency combinations (specified by the customer requisition).

220 VAC ±10 percent, 50Hz ±2Hz, three phase.

208 VAC ±10 percent, 60Hz ±2Hz, three phase.

2.21 Optional equipment. When specified by the customer requisition, the following equipment shall be supplied with the freezer:

a. Remote alarm indicators.
b. A liquid nitrogen back-up cooling system (if desired by customer).
c. Storage rack system to facilitate inventory requirements.

2.22 UL 471. The freezers shall conform to UL 471, Standard for Commercial Refrigerator and Freezers. Acceptable evidence of this shall be a UL listing mark or a certified test report from an independent laboratory acceptable to the Government.

2.23 Dimensions and weight. Exterior dimensions of the freezer shall not exceed 111" long x 41" front to back x 54" high. Maximum weight shall not exceed 2,000 pounds empty.

2.24 Identification marking. The freezers shall have a readily accessible, permanently mounted label with at least the following information (two labels may be provided if they are mounted adjacent to one another):

a. Item identification
b. Manufacturer's name and Government-assigned CAGE
c. Contract number and date of manufacture
d. Freezer's serial number
e. National Stock Number (NSN)
f. Electrical requirements
g. Maximum sustained operating current draw
h. Locked compressor rotor current draw
i. Low and high-side design pressures (in accordance with UL 471)
j. Refrigerant type and amount required (in accordance with UL 471)



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k. Maximum current draw of battery charger
l. Product intended for indoor use

2.25 Wiring diagram. The wiring system diagram shall be located so that it will be visible when power supply connecting to the product are being made by the user.

2.26 Manuals.

2.26.1 The contractor shall furnish manuals, handbooks and/or brochures containing complete operation, installation and service/maintenance instructions (including pictures of illustrations, as necessary) with complete schematics and wiring diagrams. All manuals, handbooks, and/or brochures shall be written in the English language and all schematics and wiring diagrams shall use American electrical and electronics symbols. The manual shall include electrical data and connection diagrams for all applicable utilities. The instructions shall also contain a complete list of all replaceable parts showing part number, name and quantity required. All service notes, service memos, etc., pertaining to the equipment and issued subsequent to the printing of the original instructions shall also be included. These instructions shall be, as a minimum, that furnished to service engineers (or servicemen) that normally install and service the equipment for the company or distributor. When the system being procured includes, or will operate ancillary equipment, the service data shall include complete instructions and drawings which show interfacing of all system components and instructions for trouble shooting, a procedure for periodic quality control testing.

2.26.2 Two complete sets of the above information shall be packaged with each unit of the required equipment. Where multiple quantities of the same unit are to be delivered to the same ultimate consignee, then only a total of two service manuals need to be supplied to each consignee. However, two operational manuals still need to be supplied with each unit.

2.27 Workmanship. The ultra low temperature, blood products freezer shall be free from defects which detract from its appearance or impair its serviceability.

2.28 Unit. One ultra low temperature, blood products freezer, as specified, constitutes one unit. Each freezer shall be supplied in a suitable commercial container.

3. REGULATORY REQUIREMENTS.



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3.1 Federal Food, Drug, and Cosmetic Act. If the product covered by this document has been determined by the U.S. Food and Drug Administration to be under its jurisdiction, the offeror/ contractor shall comply, and be responsible for compliance by its subcontractors/suppliers, with the requirements of the Federal Food, Drug and Cosmetic Act, as amended, and regulations promulgated thereunder. In addition, the offeror/contractor shall comply, and be responsible for compliance by its subcontractors/suppliers, with the requirements of all other applicable Federal, State, and local statutes, ordinances, and regulations.

3.2 Recovered materials. The offeror/contractor is encouraged to use recovered materials in accordance with the Federal Acquisition Regulation (FAR) Subpart 23.403 to the maximum extent practical.

4. QUALITY ASSURANCE PROVISIONS.

4.1 Product Conformance. The products provided shall meet the salient characteristics of this commercial item description, conform to the producer's own drawings, specifications, standards, and quality assurance practices, and be the same product offered for sale in the commercial market. The government reserves the right to require proof of such conformance.

4.2 Metric products. Products manufactured to metric dimensions will be considered on an equal basis with those manufactured using inch-pound units. Dimensions and tolerances specified shall be converted using conversion tables contained in the latest revision of ASTM E 380.

4.4 First article. Prior to the start of the contract, samples of the completed item shall be tested by the contractor. The samples shall be submitted, together with the contractor's inspection and test reports, to the procuring contracting officer for examination and testing to determine compliance with the requirements of the specification. Written approval of the materials, construction, and performance of the sample shall be obtained from the procuring contracting officer, or his authorized representative, prior to the manufacture and assembly of the remaining quantity to be manufactured. Pending approval of the first articles, acquisition of materials, components or commencement of production is at the sole risk of the contractor. Accepted sample shall become the property of the procuring activity and shall be included in the quantity called for in the contract or order. In the event that the sample is not approved,



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the contractor shall submit a new sample for approval or modify the existing sample to remove the deficiencies cited.

4.5 Tests.

4.5.1 Tests in accordance with UL 471. UL certification shall be accepted as conforming with UL 471.

4.5.2 Performance testing. The freezer shall be tested for compliance with performance requirements as specified under the Salient Characteristics paragraphs. The operation of the anti-icing capabilities (three 24 hour tests, each at a different load capacity) shall be tested prior to testing of reserve capacity. Unless otherwise specified, timing of testing shall commence when environmental conditions have been stabilized for at least one hour with the unit in operation.

4.5.2.1 The freezer shall be stored in a room maintained at 75°F with its lid open for a minimum of 24 hours in order to stabilize the chamber temperature. Temperature sensor probes shall be placed at the top, bottom and center of the front and back panels (1 inch in from left and right sides). Probes shall also be placed in center of all four panels and the bottom. The temperature and current draw shall be recorded during pull-down and for a minimum of 4 hours afterward. The freezer indicator temperature shall not differ by more than 5°F from the add-on probes during the entire test period. Each cascade refrigeration system shall be capable of pull down of an empty chamber from 72°F to -121°F in less than four hours and of maintaining the chamber at -121°F. Both cascade systems operating simultaneously shall be capable of pull down from 72°F to -121°F in less than two hours. Current draw shall not exceed the manufacturer's published data. Battery charger current draw data shall also be recorded.

4.5.2.2 While ambient temperature is between 70°F and 80°F and freezer temperature stabilized at -121°F, the empty freezer shall cycle at least once every 1/2 hour during a 4 hour period and temperature excursions shall not exceed 11°F.

4.5.2.3 With ambient temperature stabilized for 24 hours at 95°F(±2°F), relative humidity between 90 and 95 percent, empty freezer, there shall be no lid lock caused by frost build-up for the 24 hour test period and the temperature shall be -116°F or colder. During this test period, the freezer lid shall be opened at each 4 hour interval for a duration of 60 seconds. Temperature may be warmer than -82°F for three minutes after closing. This test shall be performed 3 times, once each for freezer empty, half full, and filled to capacity. Standard



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approved frozen red blood cell freezing containers filled with a fluid that mimics the heat transfer and capacity of red blood cell solutions shall be used. The vendor shall supply these test materials.

4.5.2.4 With conditions stabilized, ambient at 95°F(±2°F) relative humidity at 90 to 95 percent, and full freezer chamber at -121°F, the unit shall maintain an internal temperature of -121°F (±5°F) for a period of 96 hours. Current draw shall not increase during the 96 hours.

4.5.2.5 The ambient air temperature shall be maintained at 95°F(±2°F). The full freezer shall be stabilized at -121°F. The line power to the freezer shall be shut off for a minimum of 24 hours, during which time the freezer lid shall be opened for 10 seconds every 3 hours. The liquid nitrogen back-up system shall maintain the temperature of the freezer at operating temperature for the entire test period. (The temperature shall be recorded at least every 15 minutes.)

4.5.2.6 The units shall be tested for remote, local, and audio/visual alarm systems which alert the operator of any abnormal temperature in the freezer or if power is disconnected. The alarm systems shall be activated when temperatures exceed the warm or cold limit set points, or power is interrupted.

4.5.2.7 The temperature recorder provided with the freezer shall be tested by recording the temperature of the freezer during all the above performance tests.

4.6 Market acceptability. The following market acceptability criteria are necessary to document the quality of the product to be provided under this Commercial Item Description.

4.6.1 The company producing the item must have been producing a product meeting the requirements of this Commercial Item Description for at least two (2) years.

4.6.2 The company must have sold 1,000 units meeting this Commercial Item Description in the commercial market place over the past two (2) years.

4.7 Other quality assurance provisions. Shall be as indicated in Section E, Inspection, of the contract and/or order.

5. PACKAGING.



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5.1 Preservation, packaging packing, labeling and marking. Shall be as specified in the contract and/or order. Specific requirements for this item are indicated in Section C, Description/Specification; general packaging and marking requirements are indicated in Section D, Packaging and Marking.

6. NOTES.

6.1 National Stock Number Data. The following National Stock Number and its respective Item Identification are covered by this document:

| NATIONAL STOCK NUMBER | ITEM IDENTIFICATION |
|---|---|
| 4110-01-234-8153 | FREEZER, MECHANICAL, BLOOD PRODUCTS |

6.2 Source of Documents:

6.2.1 Government Documents

Defense Printing Service Detachment Office
Bldg. 4D (NPM-DODSSP)
700 Robbins Avenue
Philadelphia, PA 19111-5094

6.2.2 Non-Government Documents.

ASTM documents.

(Applications for copies should be addressed to the American Society for Testing and Materials (ASTM), 100 Barr Harbor Drive, West Conshohocken, PA 19428-2959.)

ANSI documents.

(Applications for copies should be addressed to the American National Standards Institute, 11 West 42nd Street, New York, NY 10036-8002.)

MILITARY INTERESTS: Preparing Activity:



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Custodians:

Army - MD
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DoD-MB

Agent:

DLA-DM

CIVIL AGENCY COORDINATING ACTIVITIES:

VA-OSS
USPHS
FDA-MPQAS
GSA-FSS-FCGC

Project Number: 4110-0519

Location: FRZ500C8.DOC (WORD 6)



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